

Log-polyhomogeneous pseudodifferential operators on filtered manifolds and their spectral asymptotics

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In this talk I will discuss the tangent groupoid theory of pseudodifferential operators whose symbols have asymptotic expansion like

$$\sigma(x, \xi) \sim \sum_{\ell=0}^k \sum_{j=0}^{\infty} a_{j,k}(x, \xi) \log(|\xi|)^{\ell}$$

as $\xi \rightarrow \infty$, where $a_{j,k}(x, \xi)$ is homogeneous of order $m - j$ in ξ .

Section 1: What is a log-polyhomogeneous function?

Thank you for listening!